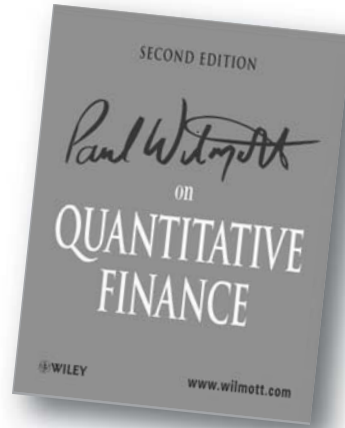


APPENDIX E

what you get if
(when) you upgrade
to *PWOQF2*



INTRODUCTION

The book you've just read is a self-contained account of 'classical' quantitative finance, at a university level. But it's not the whole story. *Paul Wilmott On Quantitative Finance, second edition*, full version, is used by 'serious' quants, in both financial institutions and academia as *the* reference for models, methods and inspiration. But it's still written in the same easy-to-digest style that you are by now familiar with :)

In the 'full' version of *PWOQF2* you'll see four types of extra material:

- More mathematical techniques, useful for valuing almost all exotic derivatives
- Beyond Black–Scholes, errors in the classical models and new models to account for them, in equity, FX, commodity markets and fixed income
- Descriptions and models for more unusual financial markets
- More advanced numerical methods

PWOQF2 is divided up into three volumes and six parts. Volume 1 contains Mathematical and Financial Foundations, Basic Theory of Derivatives, Risk and Return. Volume 2 contains Exotic Contracts and Path Dependency, Fixed-income Modeling and Derivatives, Credit Risk. Volume 3 contains Advanced Topics, Numerical Methods and Programs.

List of chapters in *Paul Wilmott On Quantitative Finance, second edition* (material not contained in the university edition is in bold).

Volume 1: Mathematical and Financial Foundations; Basic Theory of Derivatives; Risk and Return

Preface

Part One: Mathematical and Financial Foundations; Basic Theory of Derivatives; Risk and Return

1. Products and Markets
2. Derivatives
3. The Random Behavior of Assets
4. Elementary Stochastic Calculus
5. The Black–Scholes Model
6. Partial Differential Equations
7. The Black-Scholes Formulæ and the Greeks
- 8. Simple Generalizations of the Black–Scholes World**
- 9. Early Exercise and American Options**
- 10. Probability Density Functions and First Exit Times**
11. Multi-asset Options
12. How to Delta Hedge
13. Fixed-income Products and Analysis: Yield, Duration and Convexity
14. Swaps
15. The Binomial Model
- 16. How Accurate is the Normal Approximation?**
17. Investment Lessons from Blackjack and Gambling
18. Portfolio Management
19. Value at Risk
20. Forecasting the Markets?
21. The Trading Game

Volume 2: Exotic Contracts and Path Dependency; Fixed Income Modeling and Derivatives; Credit Risk

Part Two: Exotic Contracts and Path Dependency

- 22. An Introduction to Exotic and Path-dependent Options
- 23. Barrier Options
- 24. Strongly Path-dependent Options**
- 25. Asian Options**
- 26. Lookback Options**
- 27. Derivatives and Stochastic Control**
- 28. Miscellaneous Exotics**
- 29. Equity and FX Term Sheets**

Part Three: Fixed Income Modeling and Derivatives

- 30. One-factor Interest Rate Modeling
- 31. Yield Curve Fitting
- 32. Interest Rate Derivatives
- 33. Convertible Bonds**
- 34. Mortgage-backed Securities**
- 35. Multi-factor Interest Rate Modeling**
- 36. Empirical Behavior of the Spot Interest Rate**
- 37. Heath, Jarrow & Morton and Brace, Gatarek & Musiela Models
- 38. Fixed-income Term Sheets**

Part Four: Credit Risk

- 39. Value of the Firm and the Risk Default**
- 40. Credit Risk
- 41. Credit Derivatives**
- 42. RiskMetrics and CreditMetrics
- 43. CrashMetrics
- 44. Derivatives **** Ups

Volume 3: Advanced Topics; Numerical Methods and Programs

Part Five: Advanced Topics

- 45. Financial Modeling**
- 46. Defects in the Black–Scholes Model**
- 47. Discrete Hedging**
- 48. Transaction Costs**
- 49. Overview of Volatility Modeling
- 50. Deterministic Volatility Surfaces**
- 51. Stochastic Volatility**
- 52. Uncertain Parameters**
- 53. Empirical Analysis of Volatility**
- 54. Stochastic Volatility and Mean-variance Analysis**
- 55. Asymptotic Analysis of Volatility**
- 56. Volatility Case Study: The Cliquet Option**
- 57. Jump Diffusion**
- 58. Crash Modeling**
- 59. Speculating With Options**
- 60. Static Hedging**
- 61. The Feedback Effect of Hedging in Illiquid Markets**
- 62. Utility Theory**
- 63. More About American Options and Related Matters**
- 64. Advanced Dividend Modeling**
- 65. Serial Autocorrelation in Returns**
- 66. Asset Allocation in Continuous Time**
- 67. Asset Allocation Under Threat of a Crash**
- 68. Interest-rate Modeling without Probabilities**
- 69. Pricing and Optimal Hedging of Derivatives, the Non-probabilistic Model Cont'd**
- 70. Extensions to the Non-probabilistic Interest-rate Model**
- 71. Modeling Inflation**
- 72. Energy Derivatives**
- 73. Real Options**

74. Life Settlements and Viaticals

75. Bonus Time

Part Six: Numerical Methods and Programs

76. Overview of Numerical Methods

77. Finite-difference Methods for One-factor Models

78. Further Finite-difference Methods for One-factor Models

79. Finite-difference methods for Two-factor Models

80. Monte Carlo Simulation

81. Numerical Integration

82. Finite-difference Programs

83. Monte Carlo Programs

Appendix: All the math You Need . . . and No More (an Executive Summary)

Bibliography

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